

Module 04: Cognition — Enactivism and 4E

Cognition

Course: Active Inference 401 | **Unit:** Philosophical Foundations | **Audience:**
Advanced undergraduates / graduate students

Learning Objectives

1. Evaluate the relationship between Active Inference and the **4E cognition** framework (Embodied, Embedded, Enacted, Extended).
2. Analyze the **enactivist challenge**: Does Active Inference honor or betray the enactivist program?
3. Examine the **extended mind thesis** through the lens of Markov Blankets.

Key Concepts

1. The 4E Framework

The "4E" movement in cognitive science challenges classical cognitivism (computation over internal representations):

- **Embodied:** Cognition depends on the body's morphology and sensorimotor capacities
- **Embedded:** Cognition exploits environmental structure (scaffolding, cognitive niche construction)
- **Enacted:** Cognition is constituted by the organism-environment coupling, not by internal computation
- **Extended:** Cognitive processes can extend beyond the brain into tools, notebooks, and social structures

Active Inference engages with all four dimensions:

- **Embodied:** The generative model necessarily includes a body model; active inference is inherently sensorimotor

- **Embedded:** Environmental regularities are captured in the generative model's priors
- **Enacted:** The agent-environment coupling through the Markov Blanket is constitutive of cognition
- **Extended:** The extended mind thesis is contested — does the Markov Blanket *prevent* or *enable* extension?

2. The Enactivist Challenge

Enactivism (Varela, Thompson, Rosch, 1991; Thompson, 2007) makes strong claims:

Radical enactivism: Cognition is *not* internal representation — it is *sense-making* through organism-environment interaction. There are no representations, only dynamic coupling.

The tension with Active Inference: Active Inference is explicitly representationalist — it involves generative *models* that *represent* (in some sense) the causal structure of the world. This creates a philosophical clash:

- **Enactivist critique:** Active Inference smuggles in the very representationalism that enactivism rejects. Calling the generative model a "model" implies representation.
- **Active Inference response:** The generative model is not a static representation but a *lived structure* — it's enacted through perception-action loops. Ramstead et al. (2020) argue "Active Inference *is* enactive inference."
- **Middle ground:** Perhaps the debate is about the word "representation." If "representation" means a detached, context-independent symbol, then Active Inference isn't representationalist. If it means any internal structure that covaries with external states, then it is.

3. The Extended Mind and Markov Blankets

Clark & Chalmers' (1998) extended mind thesis: cognitive processes can extend beyond the skull (Otto's notebook is part of his memory).

The Markov Blanket paradox: If the MB defines the boundary of the cognitive system, doesn't it *prevent* extension by construction? The MB separates internal from external states — how can cognition extend beyond it?

Responses:

- **Nested blankets:** The MB is scale-dependent. At one scale, the brain has a blanket. At another scale, the brain-notebook system has its own blanket. Extension happens through nesting.
- **Cognitive niche construction:** The organism reshapes its environment (building Markov Blankets into the world) — this is the formal version of "extended cognition."
- **Cultural Markov Blankets:** Social institutions, languages, and technologies create group-level statistical boundaries that support collective cognition.

4. Autopoiesis and the FEP

Autopoiesis (Maturana & Varela): Living systems are self-producing — they maintain their own organization through their own processes.

The FEP's relationship to autopoiesis:

- **Convergence:** Both emphasize self-organization, operational closure, and the fundamental circularity of living systems
- **Divergence:** Autopoiesis rejects representationalism; the FEP embraces generative models
- **Synthesis:** The FEP can be seen as a mathematical formalization of autopoietic principles — self-organization as free energy minimization

5. Cognitive Niche Construction

Active Inference naturally explains **cognitive niche construction**: organisms don't just adapt to their environment — they *modify* their environment to reduce future surprise:

- Building shelter reduces prediction error about temperature
- Creating written language extends the generative model into external media
- Developing institutions creates shared predictive structures

This connects to Sterelny's (2010) "scaffolded minds" and Heyes' (2018) "cognitive gadgets."

Summary

Active Inference engages deeply with the 4E cognition framework but is contested by radical enactivists who reject its representationalism. The extended mind thesis creates a productive tension with Markov Blankets. Cognitive niche construction provides a formal account of how organisms shape their environments to support inference.

Further Reading

- Ramstead, M. J. D. et al. (2020). A tale of two densities: Active inference is enactive inference. *Adaptive Behavior*, 28(4), 225-239.
- Clark, A. & Chalmers, D. J. (1998). The extended mind. *Analysis*, 58(1), 7-19.
- Thompson, E. (2007). *Mind in Life*. Harvard University Press.
- Constant, A. et al. (2019). A variational approach to niche construction. *Journal of the Royal Society Interface*, 15, 20170685.